

of traditional animators months to create by hand was produced 'automatically'.

Designing sequences with behavioural rules is not limited to character animation. For *Independence Day* (1996), some of the sequences of dog-fighting spaceships and F-18 aircraft contained many hundreds of individual ships. To animate these sequences by hand would have been impossibly time-consuming. Instead, software engineers programmed the computer model ships so that they knew how to fly – spacecraft and aircraft each being able to achieve different speeds and manoeuvres. The craft were then given a range of objectives and priorities; spacecraft were instructed to attack the nearest aircraft unless another aircraft entered their own defensive zone. Each type of craft was given different types of weapon and instructed which other craft it could fire at and what range was needed to do so. When a craft was hit by a missile, the computer

Mathematical behavioural rules determined the animation of complicated air battles in Independence Day (1996).

was programmed to swap the model of the flying craft for a model of a disintegrating craft. The luck of aliens and humans was also



Particle systems were used to generate the stunning tornadoes in Twister (1996)

affected by a simple control. At the start of the fight, the aliens were programmed to be more successful, but by the end, the US air force was given the upper hand.

Procedural animation can also be used to animate materials such as cloth, whose characteristics are difficult to replicate using normal animation techniques. By altering the parameters of a mathematical cloth algorithm, the fabric can be made to flap like silk or undulate like the canvas on a ship's mast.

A variation of this type of animation can be used to create large volumes of moving material, such as dust, snow, smoke or even plagues of locusts. 'Particle systems' deal with huge numbers of objects whose movement would be impossible to control in any other way. To use a particle system, a computer animator typically defines one or a number of 'emitters' – that is, the points from which particles will emerge on screen. It is then possible to define virtually any characteristic for the particles. The animator can select how many particles to have, how fast they are emitted, how far and fast they travel, how quickly they disperse, how they are affected by wind and gravity, whether or not they bump into one another and what happens when they do, and so on. Once the flow of the particles is satisfactory, the particles are replaced with the objects required – perhaps computer-generated fish that already have their own movements.

Performance capture

Since the dawn of motion picture photography and animation artists have sought ways to capture and reproduce naturalistic movement. Eadweard Muybridge (<9) developed a photographic



system that could capture the individual poses of moving creatures, and Max Fleischer designed the rotoscope (<54) so that animators could copy human motion accurately for use in cartoons. Today, most computer animation is achieved using a mouse to manipulate digital models and set key-frames within the digital environment. However, the idea of harnessing real movement is still attractive to film makers, and a number of methods of channelling externally driven movement into the computer have been developed.

Initial plans for creating the dinosaurs in *Jurassic Park* (1993) involved building full-scale robotic dinosaurs and traditional stop-motion puppets that would be animated by Phil Tippett. Tippett's hand-animated dinosaurs were to be digitally composited (<78) into real landscapes by Industrial Light and Magic (ILM), who planned to give them added realism by applying digital motion blur (183>) to their movements.

Digital technology was not initially considered capable of creating convincing close-up dinosaurs, but early tests by ILM convinced Steven Spielberg of the potential of the computer. As a result, the task of creating fifty dinosaur shots was moved away from the traditional method of stop-motion and into the relatively unexplored territory of the computer.

Tippett was devastated that the computer had made its first serious incursion into the world of the stop-motion animator. Like many other artists, Tippett feared that his skills would soon become redundant. However, while ILM's computer wizardry was capable of manufacturing convincing dinosaurs, the technology was not yet capable of producing a good performance.

'We realized that there had to be a way to tap the experience and art of stop-motion animators and feed that into the computer,' says Tippett. 'By combining the two disciplines, we thought pretty amazing things ought to be possible.' In conjunction with ILM, Tippett and his team built an interface between computer and traditional animator that they called the DID (dinosaur input device, later refined to direct input device).

'The DID was basically a traditional stop-motion armature,' states Tippett, 'but the DID was different in that instead of being covered with foam latex and painted to look like a dinosaur, it remained just a skeleton. Each joint of the skeleton had a number of encoders that could measure the degree and direction of movement at that pivot point.' The DID device was electronically linked to the computer, where a wire-frame digital model of the dinosaur had been built with corresponding joints. Using the DID, stop-motion artists animated a dinosaur's performance in the traditional way, then saw that movement translated to the digital model within the computer environment.

The process combined the best of both traditional and cutting-edge techniques. Animators were able to manipulate the armature to produce a sensitive performance but, unlike the old method of capturing movement on film, the new method captured digital movement instructions and allowed the animator to go back and subtly change any moves which they were unhappy with. After

Like many prominent contemporary American animators, John Lasseter (1957-) was a graduate of the Disney Character Animation programme run by the California Institute of the Arts in the 1970s.

While working on Disney features such as *The Fox and the Hound* (1981), Lasseter saw the early computer animation being created for another Disney production, *TRON* (1982), and became fascinated by the potential of the computer. Lasseter then produced his own short film based on Maurice Sendak's book *Where The Wild Things Are*, showing how traditional hand-drawn animation could be combined with computerized camera movements and environments.

In 1984 Lasseter left Disney to join the computer graphics unit of Lucasfilm, run by Dr Edwin Catmull. There he worked on a number of 3-D computer-animated projects including the short film *The Adventures of Andre and Wally B* (1984). He also animated the CG stained-glass knight for *Young Sherlock Holmes* (1985).

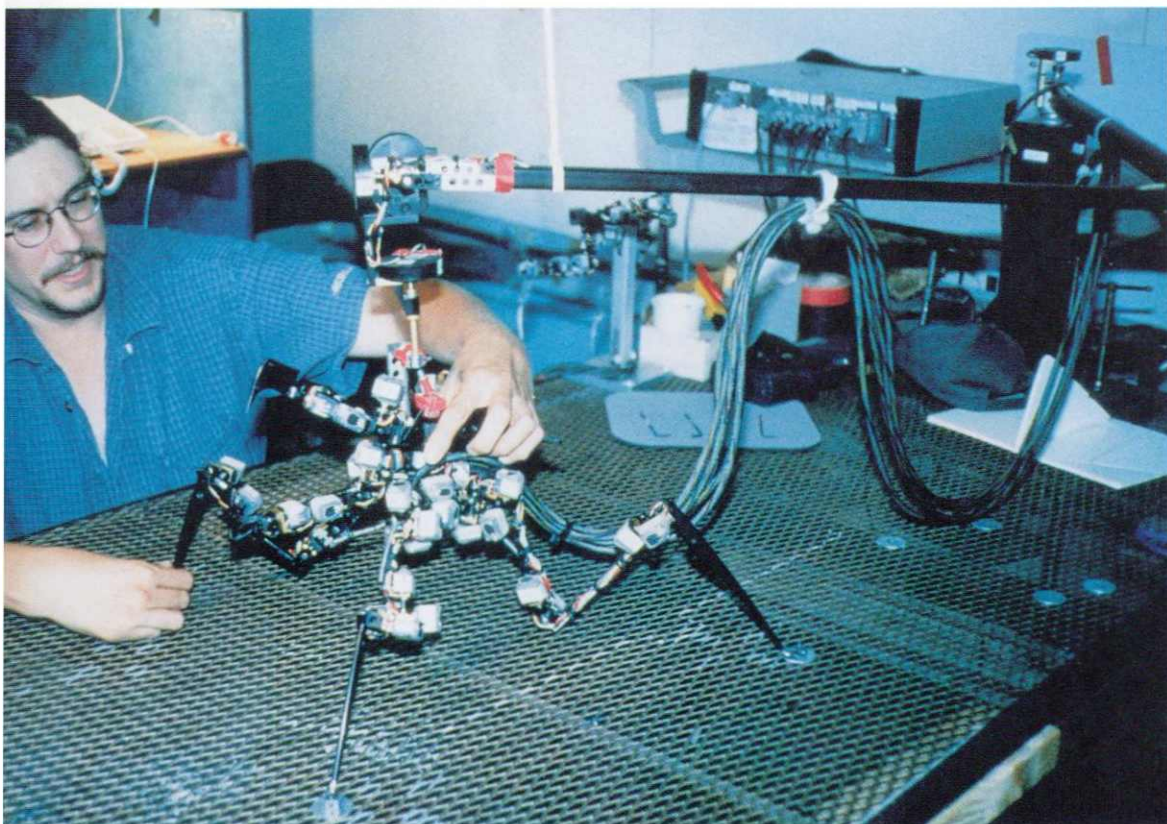
In 1986 Catmull and Lasseter formed an independent animation studio called Pixar and began production of a number of ground-breaking short films including *Luxo Jnr* (1986) and *Tin Toy* (1988), which became the first computer-animated film to win an Oscar.

Lasseter became widely praised for his extraordinary skill in creating strong characters from expressionless and inanimate objects. In the popular *Luxo Jnr*, he created a believable father and son relationship between a pair of animated desk lamps in the space of just 90 seconds.

In 1991 Pixar signed a three-picture deal with Disney. The result was *Toy Story* (1995) - the world's first entirely computer-animated feature film. *Toy Story* was a huge commercial and critical success. A number of other fully computer-animated films have been produced in the years since *Toy Story's* release, but with the production of *A Bug's Life* (1998) and *Toy Story 2* (2000), John Lasseter has continued to pioneer the medium he helped to create.



JOHN LASSETER



A digital input device built by Tippet Studio for *Starship Troopers* (1997).

basic animation was approved, ILM added the digital muscle and skin to create lifelike animals. The DID device was used to create the terrifying tyrannosaurus attack on the Jurassic Park jeeps, and the sequence in which two children are stalked by velociraptors in the kitchen.

Despite the advances in computer animation techniques, and the growing experience and skill of computer animators, the Tippet Studio continues to use DID technology to fuse computer graphics with traditional animation. The result is some of the most convincing creature performances ever created, such as that of the stunning insect stars of *Starship Troopers* (1997).

To create convincing human motion within the digital environment, another method of capturing movement has been perfected. In the 1980s, various medical establishments created methods of digitally recording and analyzing the movement of patients who suffered from leg and spinal problems. This technology has been harnessed and developed for use in movie special effects production.

'There are two main methods of capturing human movement for use in computer animation,' explains Tom Tolls of House of Moves, an Californian company that specializes in capturing movement data for use by special effects facilities. 'Optical motion capture uses a number of cameras to "film" movement and translate it into computer data [fig. 6]. Magnetic motion capture, on the other hand, uses a strong magnetic field to measure the movements of a person dressed in a body suit that is covered with magnetically active markers. Magnetic motion capture is generally more restrictive and complicated to use, but it does produce excellent data that can be

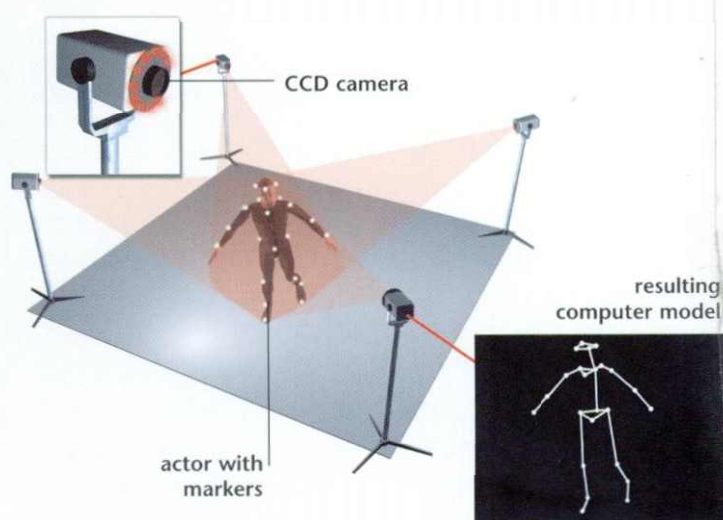
utilized to create real-time performances. This means that a performer's actions can instantly affect the movement of a computer character in live-broadcast applications such as kids TV, where animated characters need to interact with an audience.'

House of Moves uses the more popular optical method to capture movement. 'Our system works by using somewhere between eight and twenty-four CCD [charge coupled device] video cameras arranged around a performance area that is perhaps 6m [20ft] in diameter,' states Tolls (fig. 6 for basic set-up). 'The performer is dressed in a black suit to which we attach a number of small plastic balls. The balls are covered in a highly reflective material similar to

Scotchlite, which was once used for front-projection purposes [69]. Depending on the sophistication of motion detail needed, performers are dressed with between twenty and fifty markers at all the key joints. For basic movement, a hand can be represented by two markers, one at the wrist and one at the tip of the hand. Really complex movement can involve having a separate marker to capture the movement of each finger.'

During the capture process, performers are filmed moving within the capture area. Each CCD video camera emits infra-red light, which hits the reflective body markers and is directed back into the camera from which it came. The result from each camera is a black image

Figure 6 PERFORMANCE CAPTURE



JURASSIC PARK

Early plans for Steven Spielberg's (1947-; <34) adaptation of Michael Crichton's novel *Jurassic Park* (published 1990) involved a combination of full-scale animatronic dinosaurs and traditional stop-motion animation techniques (<150 and 230>). However, in the wake of their success with digital characters in *Terminator 2* (1991), the artists at Industrial Light and Magic were able to convince Spielberg that digital dinosaurs were a viable alternative to stop-motion.

To create their digital dinosaurs, the artists at ILM, under the guidance of special effects guru Dennis Muren, employed a number of groundbreaking new computer techniques. The dinosaurs themselves were first created as clay models and then cyberscanned (<162) to produce digital models. The digital models were then given a skeleton so that they could be animated using inverse kinematics (<166). ILM software engineers developed a program called 'Envelope', which allowed the skin of the model to move freely over internal muscles in order to create the look of a real animal.

ILM also created a program called 'Viewpaint', which allowed artists to paint the surface details of a dinosaur's skin directly on to the digital model. These are known as texture maps (<129) and are normally painted as a flat design that is then wrapped around the 3-D digital model, but this can cause stretching and distortion of the design. Viewpaint allowed designs to be painted in their correct place on the model itself. Several layers of texture map were applied to each dinosaur model, including bump maps to describe the contours of the skin and additional maps to describe skin colour and pattern, mud and dirt. For the sequence in which the T-Rex attacks Jurassic Park's jeeps in the rain, a special shader was written to create the effect of water running down the creature's back.

Computer animators created the dinosaur performances by referring to simple animatics – low-resolution tests that were created and refined using traditional stop-motion puppets or basic computer models. The dinosaurs were then animated using a combination of key-frame techniques (<166) and a 'dinosaur input

device' (<168), which allowed traditional stop-motion animators to transfer their animated performances directly into the computer.

Once dinosaurs had been animated and rendered (182>), they were composited (<78) into live-action background plates. The compositing process was vital to the credibility of the dinosaurs – even convincing digital dinosaurs will look fake if they do not interact with their environment. As the T-Rex ran, small clouds of dust and dirt were added beneath its feet, and splashes of water were placed where it stepped in puddles. For shots in which dinosaurs emerged from behind objects, each frame of the plate was rotoscoped by hand (<54) so that the digital creatures could be sandwiched within the scene.

As production progressed, ILM became more confident in its ability to deliver photo-realistic dinosaurs, and the creatures were given a greater on-screen presence. A late addition to the schedule was a scene in which the actor Martin Ferrero is eaten by the T-Rex. According to the original plan, the character would be seen disappearing out of frame when snatched from his seat on a toilet. The revised version showed the actor being picked up, tossed around and then eaten by the dinosaur in a single shot. During compositing, the real performer was swapped for a digital character

just as the dinosaur's mouth closed over his body. Though a last-minute decision, the shot became a show-stopper.

During the planning stages, digital dinosaurs had only been intended for use in long-shots. However, as ILM's digital work became increasingly convincing, the creatures were brought closer to the camera. For the last sequence featuring the tyrannosaurus rex, the dinosaur's texture maps were repainted with extra

detail and the creature was animated as if moving just a few metres or feet in front of the camera as it rampaged through the Jurassic Park visitor centre.

On its release the public flocked to see *Jurassic Park*. The film smashed box-office records worldwide to become the highest grossing film in history and later won Academy Awards for its special effects and sound effects. Perhaps the film's greatest legacy was to prove that, finally, the computer could help special effects artists create almost anything the scriptwriter could conceive. Film making was changed forever.





A stunt performer covered with tracking markers slides down a tilted deck, providing information for the creation of a computer-generated victim in Titanic (1997).

containing a number of moving white dots – a 2-D representation of movement from a single position. Seeing this cloud of floating points is a strange experience. Although nothing connects the markers, their movement fools the mind into connecting dots to form a sense of human movement.

To extract useful information, the computer to which this data is fed compares the position of each dot from several camera angles and calculates its exact position in 3-D space. 'The computer needs to cross-reference continually between a number of camera angles,' states Tolls. 'Looking at a performer from just one or two angles isn't enough, because each moving marker is continually being hidden behind various parts of the body.' Although the system is designed to calculate the position of each marker automatically, the technology is not foolproof and the resulting data has to be 'cleaned up' manually.

'Motion capture data has to be organized before it can be used,' explains Tolls. 'The markers are continually moving about. Sometimes one can pass another, and then the computer can get confused and swap them. The result could be an arm movement getting mixed with a hip movement, which would result in a pretty weird-looking performance!'

'This system produces a huge amount of data,' notes Tolls. 'Each CCD captures images at least sixty times a second, and with thirty markers on a body, that's 1,800 measurements per second per camera. However, it is because the system is so sensitive that it can capture recognizable performances. We do a lot of capture for the games industry, which hires famous sports stars to come and have their basketball or boxing

Computer-generated performers stroll on the decks of the Titanic.

The ship is actually a 45-foot model, while the water, seagulls and even the flag on the ship's stern are digital.



movements captured for computer games characters. We've captured the surprisingly distinctive moves of Bruce Willis for the action game *Apocalypse*, and of tennis player Andre Agassi for a Nike commercial, but probably our most well-known work was for the movie *Titanic* [1997].'

'We wanted to populate the decks of the *Titanic* with convincing humans in period costume,' explains Digital Domain effects supervisor Robert Legato, who was rewarded with an Academy Award for his work on the film. 'We could have filmed real people with motion control and then composited them on to the deck of our model ship, but Jim [Cameron] wanted to do some extreme camera moves over the model that could never have been matched with motion control. By capturing the performance of real people, we could build digital model actors and give them convincing human movement. Once a movement has been captured, it exists within 3-D digital space, so the model person that it is applied to can be placed anywhere and be looked at from any angle. This way we could populate the deck of the ship with moving people that had been created in the computer to match the motion control shots of our real model *Titanic*.'

During motion capture sessions at House of Moves, Legato dressed performers in costumes that matched the shape and form of those that *Titanic* passengers would have worn. 'We came up with the idea of attaching motion control markers to the clothes and hats that people wore,' explains Legato. 'In this case, it wasn't too important to capture the internal skeletal movement of the performers. What we really needed was their gross movement – the dynamics of cloth, the way women's skirts swayed when they walked. The women who walk on the deck of the *Titanic* don't actually have any legs, just a digital dress that moves as if there were legs underneath.'

During capture, Legato directed performers to build up a library of different movements. 'We tried to get different types of movement for different characters,' explains Legato. 'We used officious strides for ship's officers, leisurely strolls for gentlemen, more relaxed walks for steerage passengers. You would think that since this method doesn't involve any facial expressions or acting, anyone could perform for it. Actually, we found that people being captured tend to exaggerate their movements and look cartoon-like. Only certain people were able to perform totally naturally.'

In some instances characters needed to do more than walk in a straight line. 'Some people needed to be walking up stairs or, in the sinking scenes, falling down the deck,' explains Legato. 'In these cases we built basic sets that the performers could interact with. The resulting digital characters were given cyberscan faces and tracked on to the deck of the model *Titanic* with stunning results.'

Motion capture has been used with great success in certain situations, but it does have its limitations. 'Motion capture is really good at getting photo-realistic human performances into the computer,' says Tom Tolls, 'but some people tend to think that it is an easy way to achieve motion for all sorts of different animated characters. Sometimes we load performers with heavy weights so that they move in a slow, ponderous way if their movement is to be used for

some kind of giant or monster. However, fantasy creatures and monsters are best created using key-frame animation techniques [*<166*], which gives them a stylized performance. The monsters of Ray Harryhausen look so fantastic because they have this mysterious way of moving.' During the production of *Godzilla* (1998), it was planned to use human motion capture to provide the movements of the stomping bipedal monster as it rampaged around New York. The results of early tests, however, showed that motion capture animation simply made the creature look like a person in a rubber suit. Motion capture was replaced by slower but more appropriate key-frame animation.

Optical motion capture can also be used to capture the subtle movements of a human face. Around forty facial markers are needed to capture the nuances of human expression. This method was used to capture the performance of actor Gary Oldman for the creation of the futuristic Doctor Smith character in *Lost in Space* (1998).

Jim Henson's Creature Shop has developed its own method of recording facial movement for digital characters. Based on its long experience of providing amazing puppet performances, Henson's has built a puppeteering system that can be used to create digital performances in real time. 'Our system is based on the technology that we use to control some of our animatronic [mechanical] creatures,' says Hal Bertram, head of Henson's digital division. 'We use a glove and a joystick, each of which is covered with potentiometers that measure the unidirectional movements made by the person controlling the device. Any one of the movements of the glove or the joystick can be set to control any aspect of movement on the digital model. The thumb and forefinger could control the movement of a character's ears, while the movement of the joystick could move its tongue. As puppeteers use the device, they can see their actions instantly influence the performance of the character on a nearby computer monitor.'

'The performance system is really a joy to use,' says Mary Victoria, an animator at Henson's. 'You can spend a little while getting to know a character and then you can really get into giving a performance. We usually capture a facial performance in several passes, building up the layers of movement each time. We might first concentrate on the ears and eyes, then in another pass we will add the lip movement and the tongue and so on. Once the performance has been captured, we can go back into the software and manually tweak parts of the performance or re-record one movement that isn't quite right. The system can be used to create performances that are full of character in a fraction of the time that key-framing takes – perhaps a couple of hours to produce a shot that might take a week with more traditional methods. It gives you a great deal of freedom as a performer.'

Fur

The creation of lifelike fur and hair has been a technical barrier ever since the advent of computer animation. Early character animation only dared to attempt creatures that had either smooth